

Suppressed Quantum Effects of Weakly Coupled Waves

Kevin Zhou



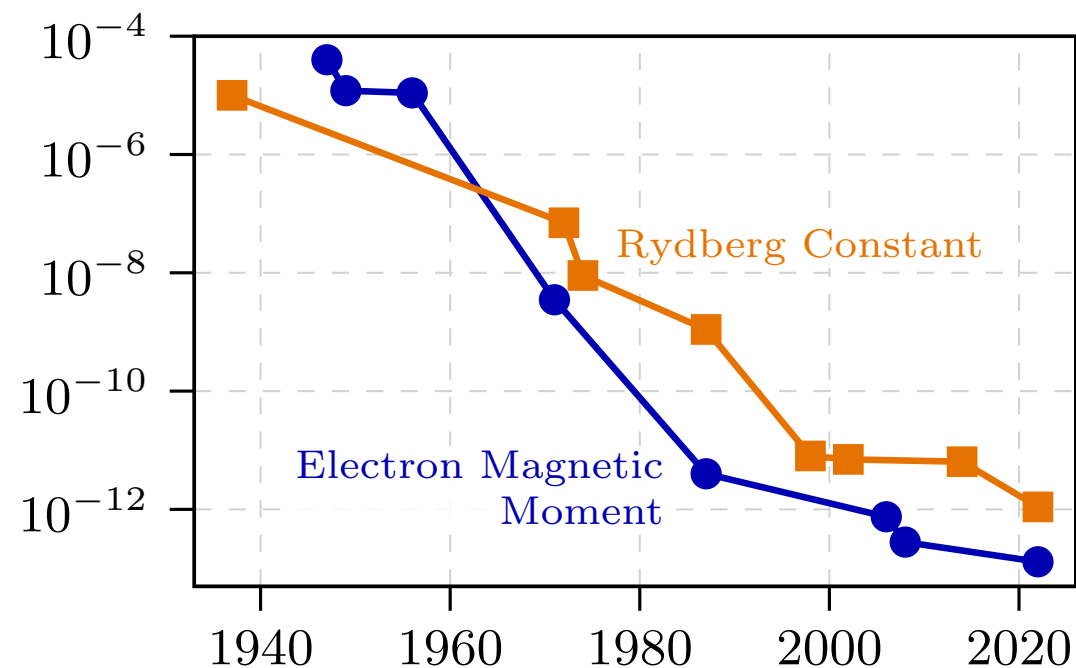
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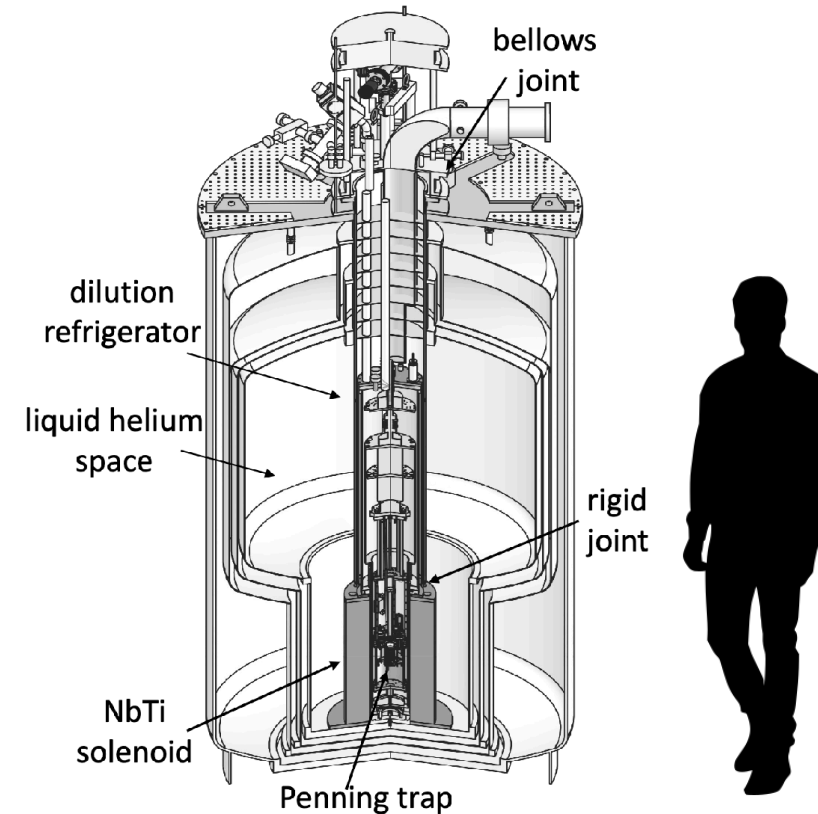
Phys. Rev. Lett. 136, 171601 (2026) and arXiv:2607.27313
with Yunjia Bao, Dhong Yeon Cheong, Nicolas Rodd, Joey Takach, Lian-Tao Wang

New Precision Experiments for Fundamental Physics

Precision measurements are improving rapidly:



Measurements of fundamental constants are 10x more precise every decade



Tabletop electron $g-2$ measurement reaches TeV-scale physics

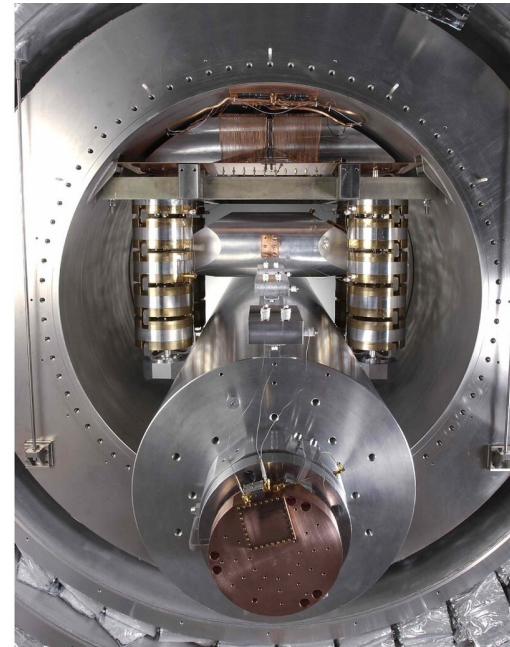
Small experiments can detect dark matter and new forces, even beyond particle colliders

My theory work focuses on maximizing the potential of this rapid experimental progress

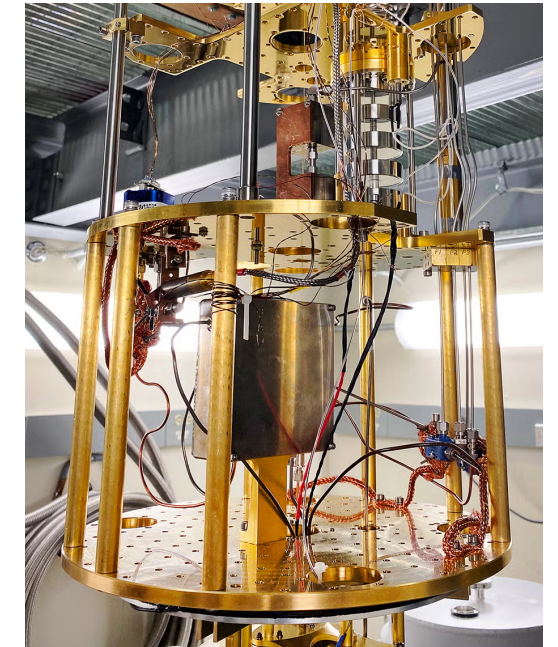
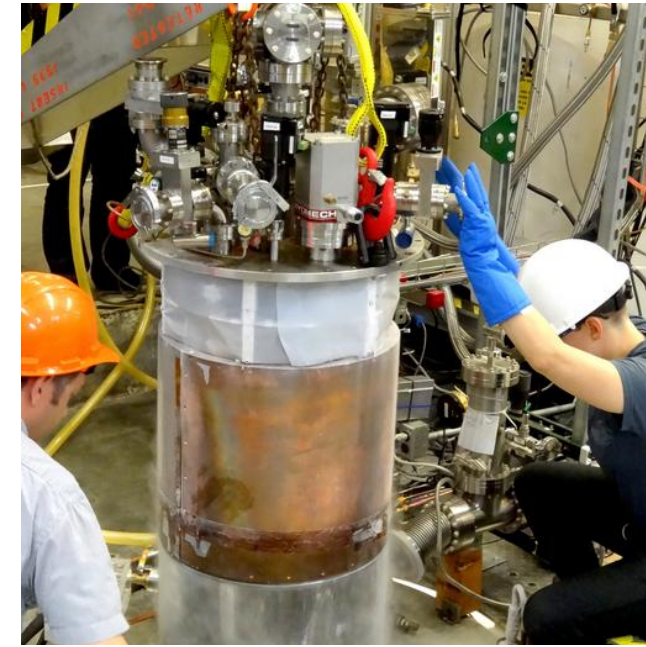
Searches for Weakly Coupled Waves

Many cutting-edge experiments search for fields with weak coupling, high occupancy

Gravitational waves: laser interferometers, Weber bars



Axion dark matter: cavity haloscopes



These detectors enhance their sensitivity by operating deep in the quantum regime

Squeezed
states

Number
measurements

Fock, “cat”, and
squeezed vacuum states

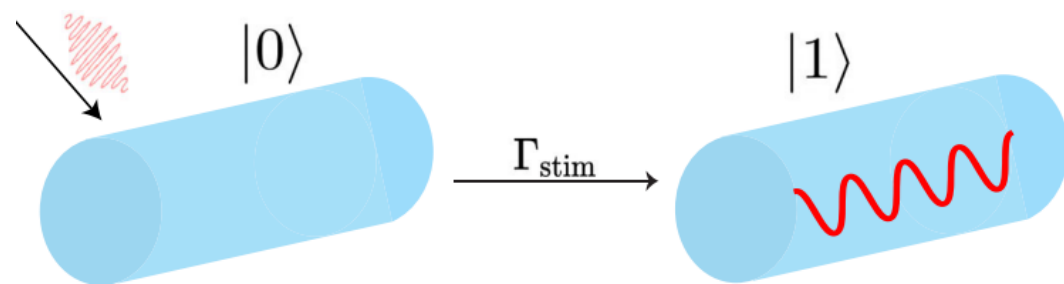
Entangled
states

Can they benefit from, or tell us about, the quantization of the field they detect?

Conflicting Conventional Wisdom

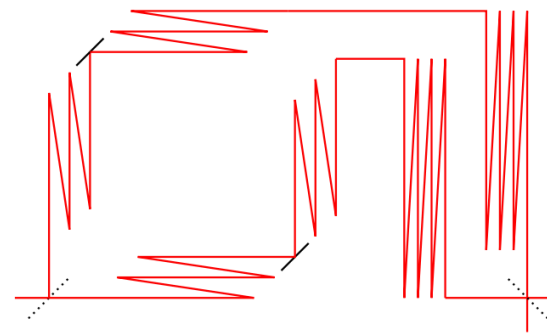
Textbook treatment describes GWs and axion DM as classical fields

But influential recent works claim quantum GW/DM effects, even probing quantum gravity!



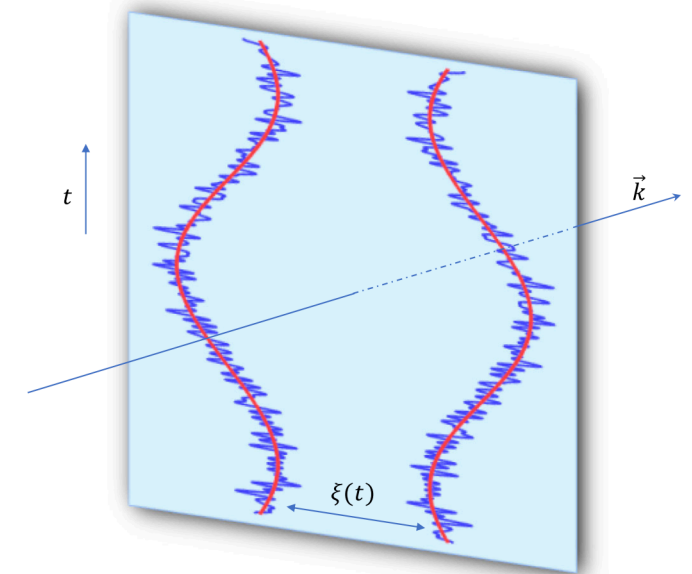
Graviton clicks

Tobar et al., Nature Commun. (2024)



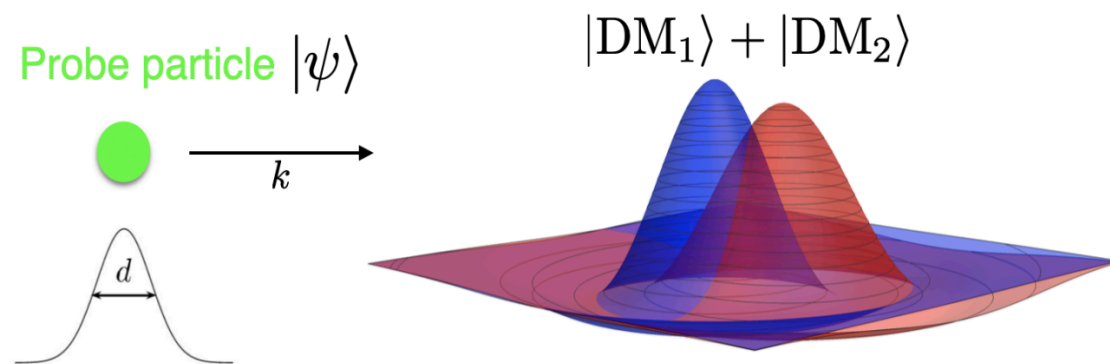
Gravitational decoherence

Schutzhold, PRL (2025)



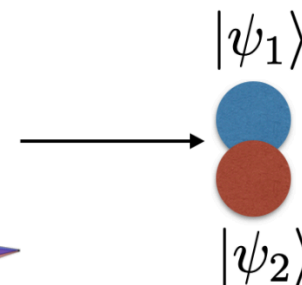
“Noise of gravitons”

Parikh, Wilczek, Zahariade, PRL (2021)



Axion cat states

Allali and Hertzberg, PRL (2021)



(+50 other recent papers with similar claims)

Defining Intrinsically Quantum Effects

Confusion arises from incompatible, incomplete definitions of “classical”/“quantum”

Quantum effects involve a quantum calculation

Must distinguish detector being quantum from the field being quantum, core insight of quantum optics

Quantum effects go away at high occupancy

Only for certain states, and claimed quantum GW/axion results all involve extremely high occupancy

Quantum effects depend on $\hbar$, vanish as $\hbar \rightarrow 0$

Depends on what you keep fixed

$$E = N\hbar\omega \rightarrow 0 \text{ as } \hbar \rightarrow 0?$$

In particle physics, only loop diagrams depend on $\hbar$
But results from quantum detectors always involve $\hbar$!

Our definition: intrinsically quantum DM/GW effects are those which cannot arise in a detector from *any* ensemble of classical fields

Outline

- **Axion DM and cavity haloscopes**

Defining the effective occupancy $N_{\text{eff}} \gg 1$ and conversion efficiency $\eta \ll 1$

GHz QCD axions and kHz GWs have $\eta_{\text{eff}} \sim 10^{-20}, 10^{-30}$ with $\eta N_{\text{eff}} \sim 1$

- **Suppression of intrinsically quantum effects**

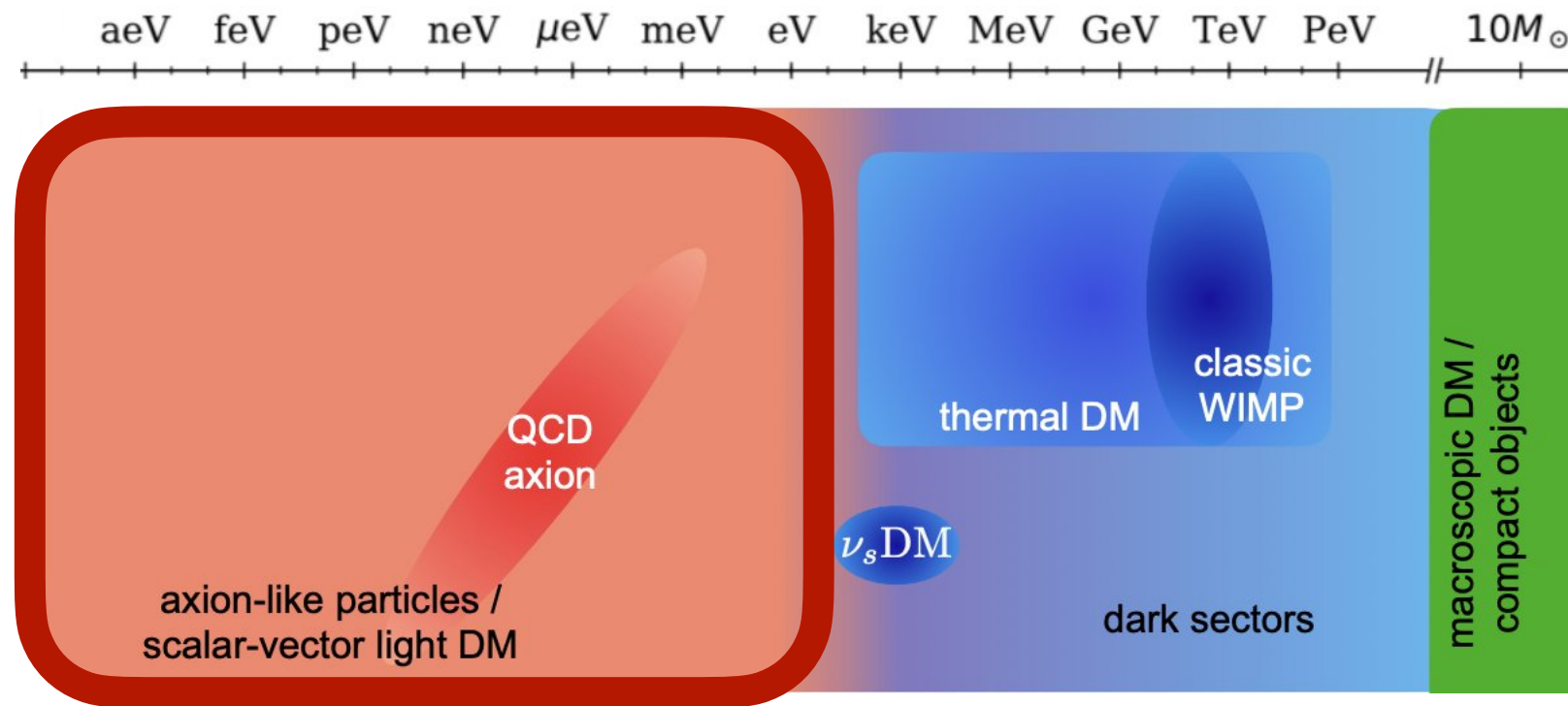
Intrinsically quantum = negativity of the Glauber P -function

Detectors only see coarse-grained modes, not fundamental modes

Intrinsically quantum effects are suppressed by an **extra** power of η

- **Addressing the literature**

Why Search for Ultralight Dark Matter?



- Simply produced: gives right amount of dark matter with minimal cosmology
- Generic: required ultralight fields automatically appear in many models
- Minimal: requires introduction of only a single new field at low energies
- Low-hanging fruit: new experiments are needed, inexpensive, and very effective
- Bounded: only a few interactions are natural and leading in effective field theory

Ultralight Dark Matter Candidates

In field theory, only a few candidates are possible!

	pseudoscalar a	scalar ϕ	vector A'_μ	axial vector V_μ
naturally light with leading coupling	✓		✓	✓
arises in high energy theories	✓✓	✓	✓	✓
can solve tuning problems	✓	?		
simply produced	✓	✓	✓?	✓?

$$(\partial_\mu a) K_{\text{EM}}^\mu$$

axion-photon

$$(\partial_\mu a) \bar{N} \gamma^\mu \gamma^5 N$$

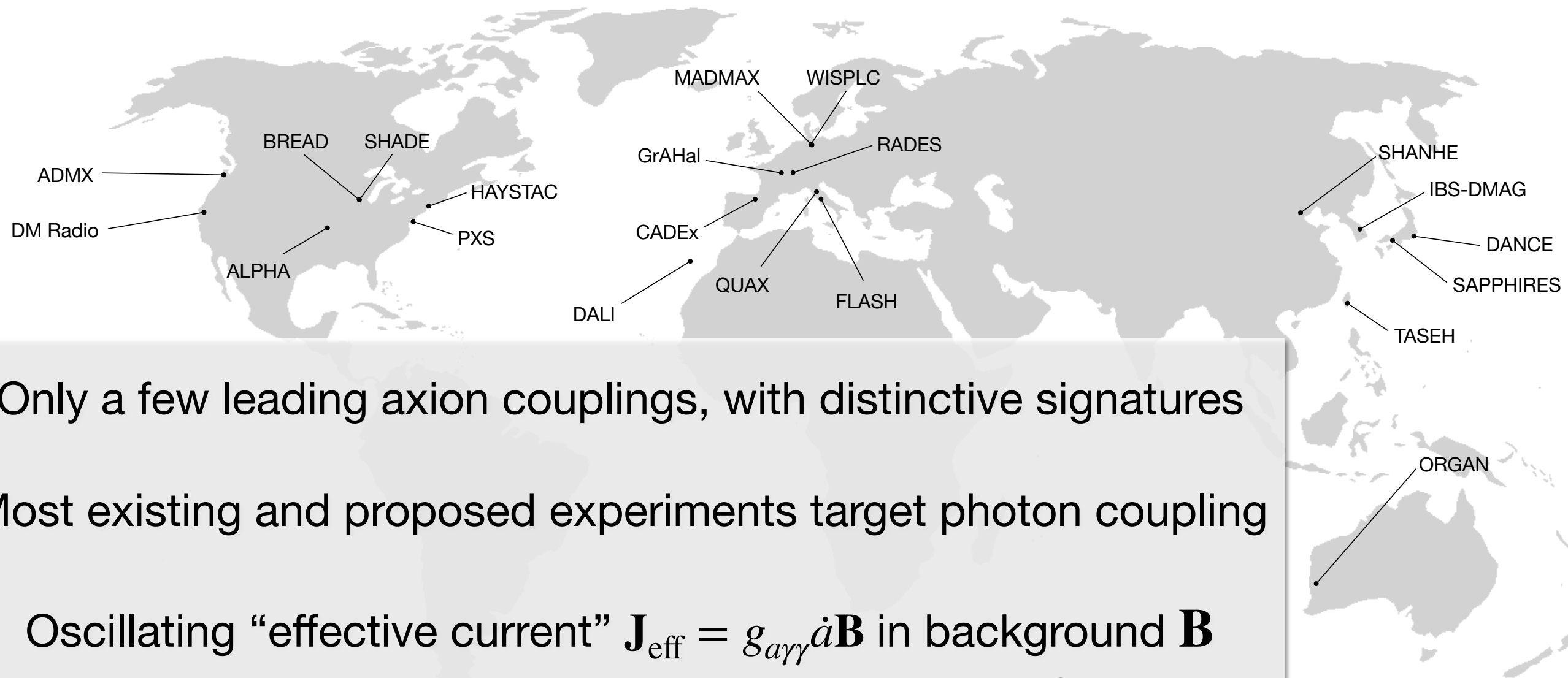
axion-nucleon

$$(\partial_\mu a) \bar{e} \gamma^\mu \gamma^5 e$$

axion-electron

$$a \bar{N} \sigma_{\mu\nu} \gamma^5 N F^{\mu\nu}$$

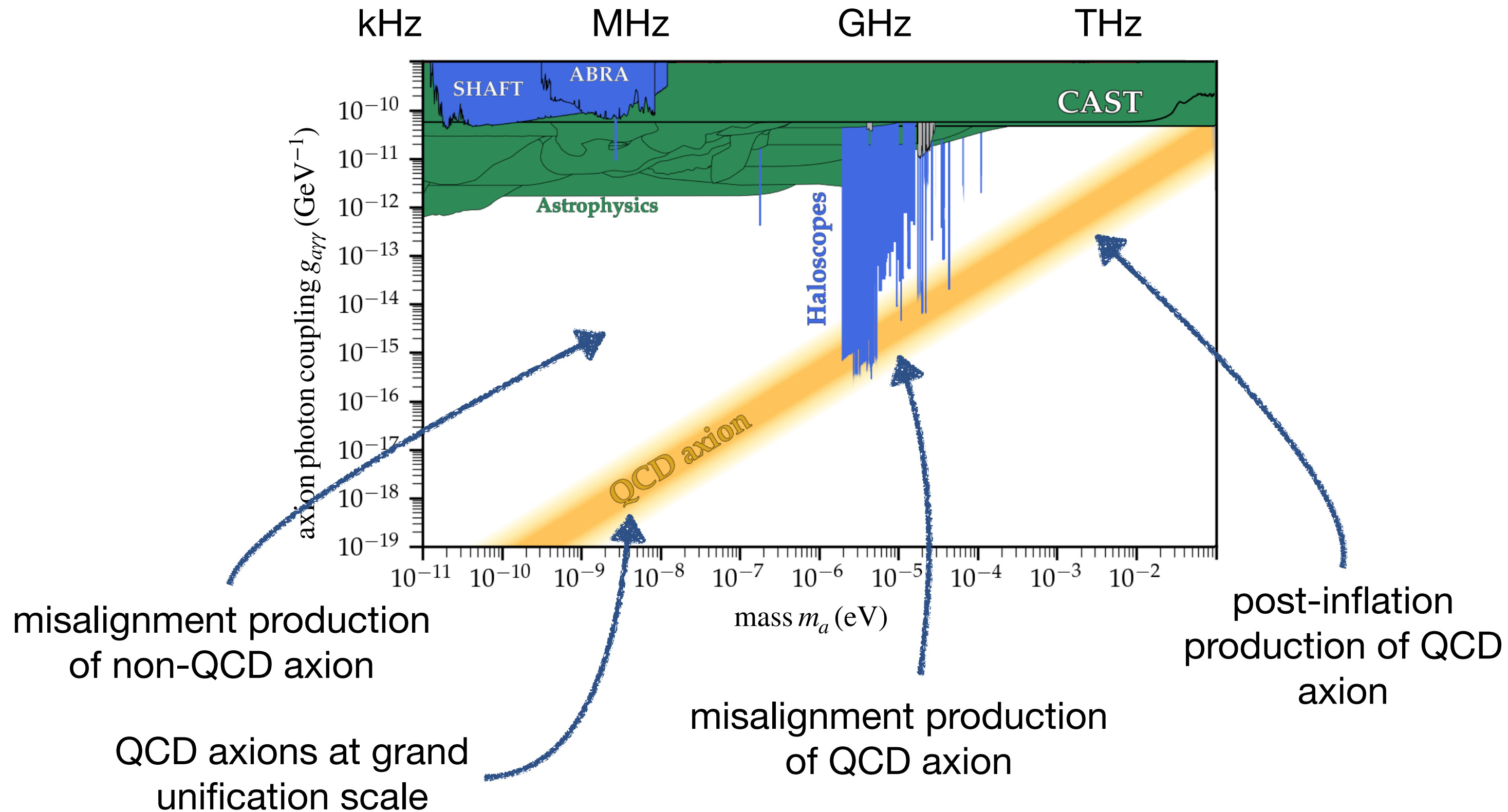
axion-EDM

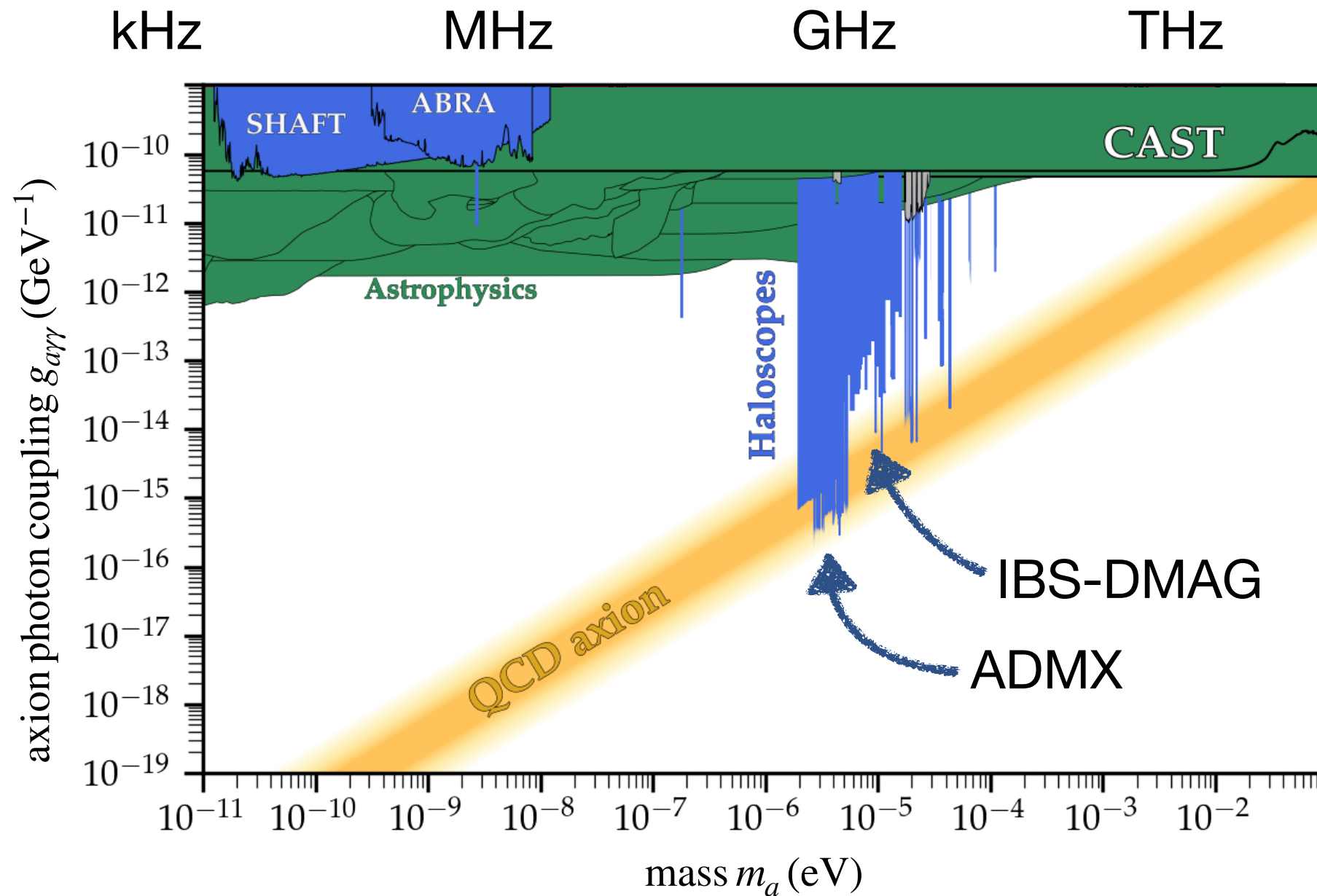


Only a few leading axion couplings, with distinctive signatures

Most existing and proposed experiments target photon coupling

Oscillating “effective current” $\mathbf{J}_{\text{eff}} = g_{a\gamma\gamma} \dot{a} \mathbf{B}$ in background $\mathbf{B}$
sources secondary, detectable electromagnetic fields

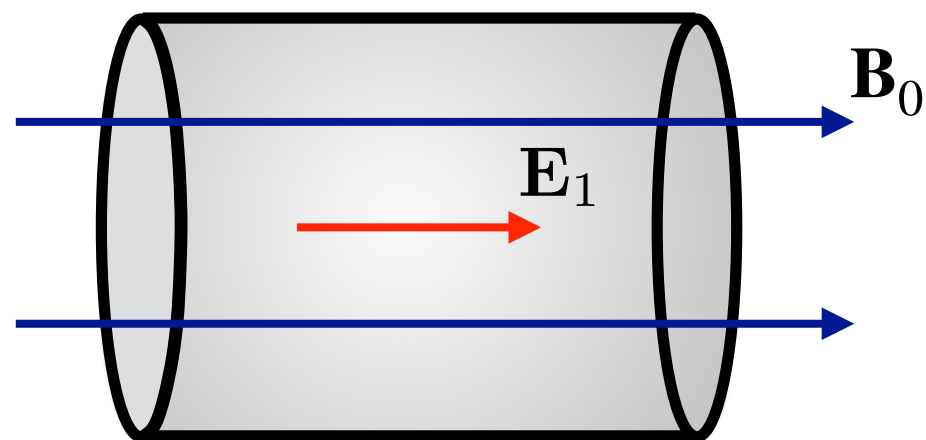




Only the axion-photon coupling has been probed significantly, and only around $f \sim \text{GHz}$, largely by two experiments using the same method

GHz Axions: The Cavity Haloscope

In background $\mathbf{B}_0$, axion drives cavity mode with profile $\mathbf{E}_1$



$$\mathbf{J}_{\text{eff}} = g_{a\gamma\gamma} \dot{a} \mathbf{B}_0$$

$$\dot{a}(t) = \sqrt{\rho_{\text{DM}}} \cos(m_a t)$$

$$P_{\text{sig}} \sim (g_{a\gamma\gamma}^2 \rho_{\text{DM}}) (B_0^2 V) (Q/m_a)$$

magnetic energy in cavity

decay time of cavity mode



Field-Free Region

Quantum Amplifier
Package

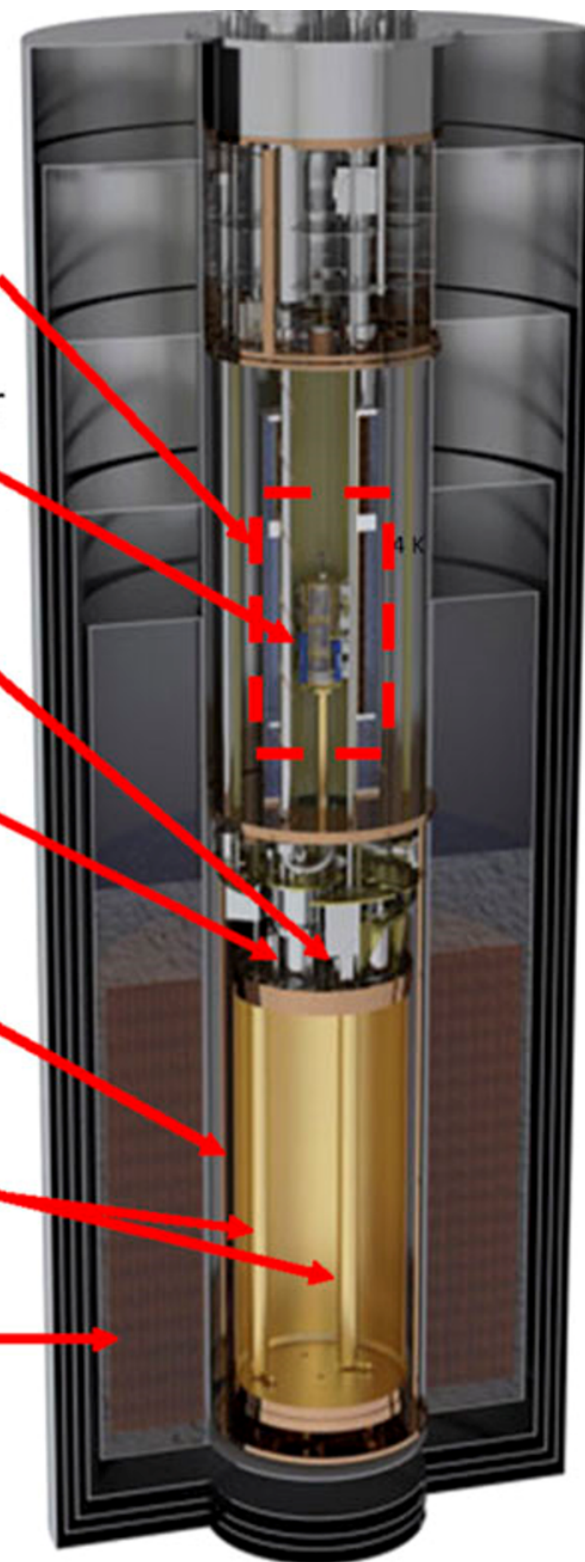
Antennas

Mixing
Chamber

Microwave
Cavity

Tuning Rods

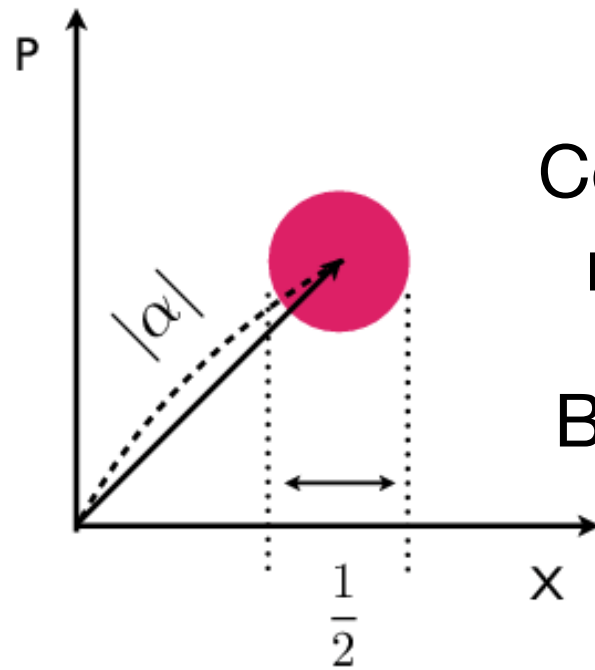
Magnet



Will build a quantum description of the axion and cavity haloscope

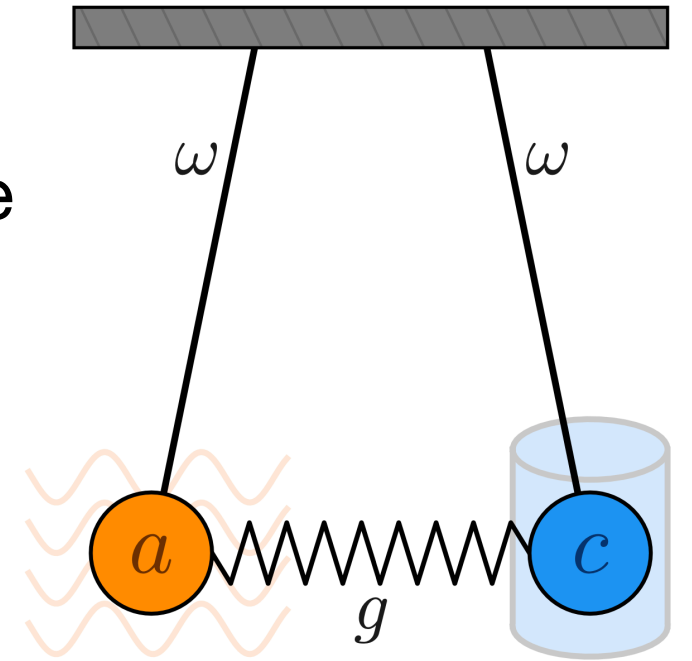
First question: what basis to use?

Coherent States Act Classically



Coherent states $|\alpha\rangle$ obey $a|\alpha\rangle = \alpha|\alpha\rangle$ and have relatively well-defined values of X , P , N , and θ

But they have a much more important property!



Axion mode and cavity mode like coupled harmonic oscillators $H_{\text{int}} = ig(c^\dagger a - c a^\dagger)$

a is a classical complex number α

cavity state $|0\rangle$ evolves to
coherent state $|\alpha gt\rangle$

quantum mode $\hat{a}$ is in coherent state $|\alpha\rangle$

joint state $|0\rangle|\alpha\rangle$ evolves to joint
coherent state $|\alpha \sin(gt)\rangle|\alpha \cos(gt)\rangle$

At leading order, coherent states act on cavity like classical field values!

Corrections to this are suppressed by $(gt)^3$, negligible for DM/GWs

Defining Nonclassical States

State of field with probability distribution $P(\alpha)$ of classical values is $\rho = \int d^2\alpha P(\alpha) |\alpha\rangle\langle\alpha|$

$$P(\alpha) = \delta^{(2)}(\alpha - \alpha_0)$$

coherent state

$$P(\alpha) \propto \delta(|\alpha| - |\alpha_0|)$$

phase-averaged
coherent state

$$P(\alpha) \propto e^{-|\alpha|^2/|\alpha_0|^2}$$

thermal Gaussian state
(default assumption for DM)

Glauber's key insight: every state corresponds to a essentially unique $P(\alpha)$!

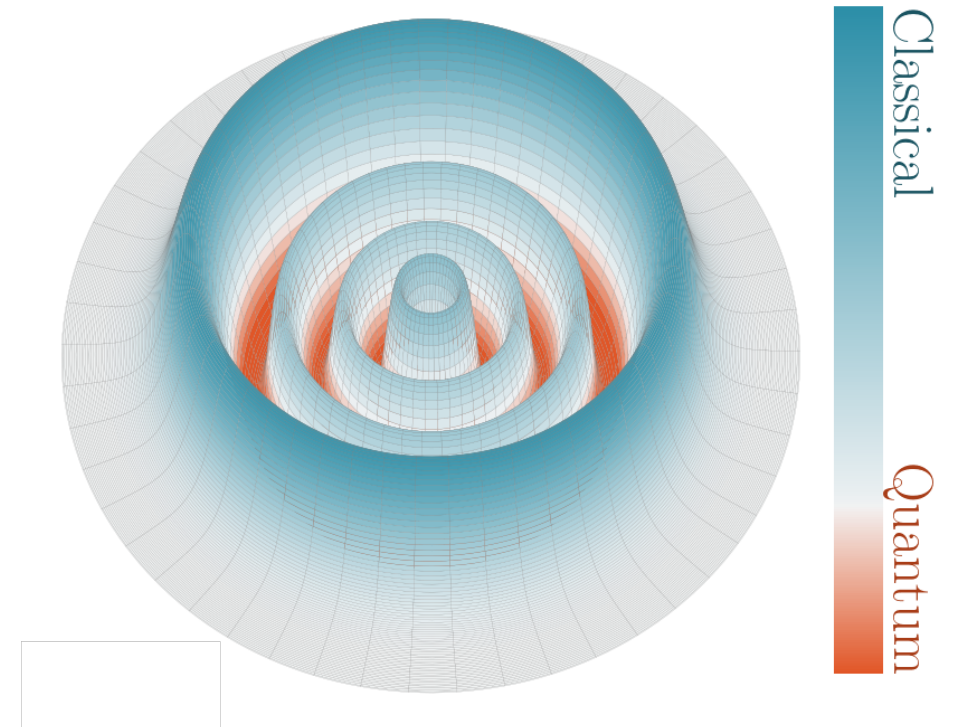
“cat” state $|\alpha\rangle + |\alpha'\rangle$

But many require $P(\alpha) < 0$, e.g. squeezed state $|\gamma\rangle$

number state $|n\rangle$

These “intrinsically quantum” states give measurement statistics that classical ensembles can't

This remains true for arbitrarily high occupancy



Outline

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- **Suppression of intrinsically quantum effects**

Intrinsically quantum = negativity of the Glauber P -function

Detectors only see coarse-grained modes, not fundamental modes

Intrinsically quantum effects are suppressed by an **extra** power of η

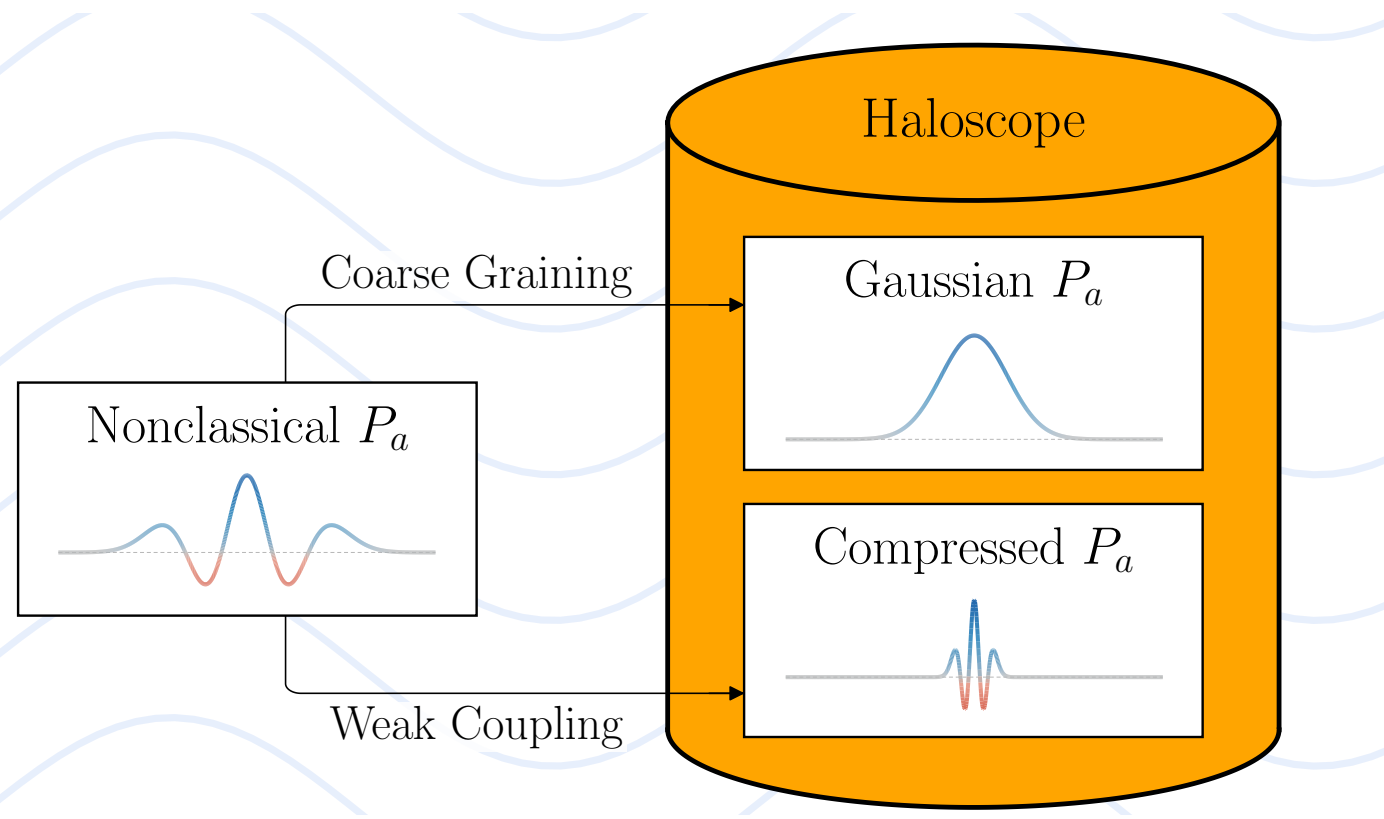
- Addressing the literature

Obstructions to Observing Intrinsically Quantum Effects

For GHz haloscope
probing the QCD axion:

Effective mode occupancy $N_{\text{eff}} \sim \rho_{\text{DM}} V / m_{\text{DM}} \sim 10^{19}$

Axion-photon conversion efficiency $\eta \sim (g_{a\gamma\gamma} B_0 Q_c / m_{\text{DM}})^2 \sim 10^{-21}$



Experiments measure coarse-grained
“effective” modes, not fundamental modes

Size of intrinsically quantum effects always
suppressed by **extra** powers of η

Quantum Description of the Cavity Haloscope

Consider coupling a cavity mode to the axion's many plane wave modes

$$H_{\text{int}} = g_{a\gamma\gamma} B_0 \int_V d^3\mathbf{x} \hat{\phi}(\mathbf{x}) \hat{E}_z(\mathbf{x}) \quad \hat{E}_z(\mathbf{x}) \supset i\hat{c}\tilde{E}_z(\mathbf{x}) + \text{h.c.} \quad \hat{\phi}(\mathbf{x}) \sim \sum_{\mathbf{p}} \hat{a}_{\mathbf{p}} e^{i\mathbf{p}\cdot\mathbf{x}} + \text{h.c.}$$

On resonance in interaction picture, reduces to a rotation between cavity mode and one “effective” axion mode

$$H_{\text{int}}(t) \simeq ig (\hat{c}^\dagger \hat{a}_{\text{eff}}(t) - \hat{c} \hat{a}_{\text{eff}}^\dagger(t))$$

Identity of this effective mode varies over the axion coherence time

$$\hat{a}_{\text{eff}}(t) = \sum_{\mathbf{p}} C_{\mathbf{p}} e^{-i(p^2/2m_a)t} \hat{a}_{\mathbf{p}}$$

Occupancy of effective mode is “mean number of axions in cavity”

$$N_{\text{eff}} = \langle \hat{a}_{\text{eff}}^\dagger \hat{a}_{\text{eff}} \rangle \sim \rho_{\text{DM}} V / m_a \sim 10^{19}$$

Effective coupling $g \sim g_{a\gamma\gamma} B_0$ implies axion-photon conversion efficiency $\eta = \sin^2(gt) \approx g^2 t^2$

Quantum State of the Effective Axion Mode

Full state of axion field has joint P -function $\rho = \int \left(\prod_i d^2\alpha_i |\alpha_i\rangle\langle\alpha_i| \right) P(\alpha_1, \alpha_2, \dots)$

P -function of the effective mode found by tracing out all other modes

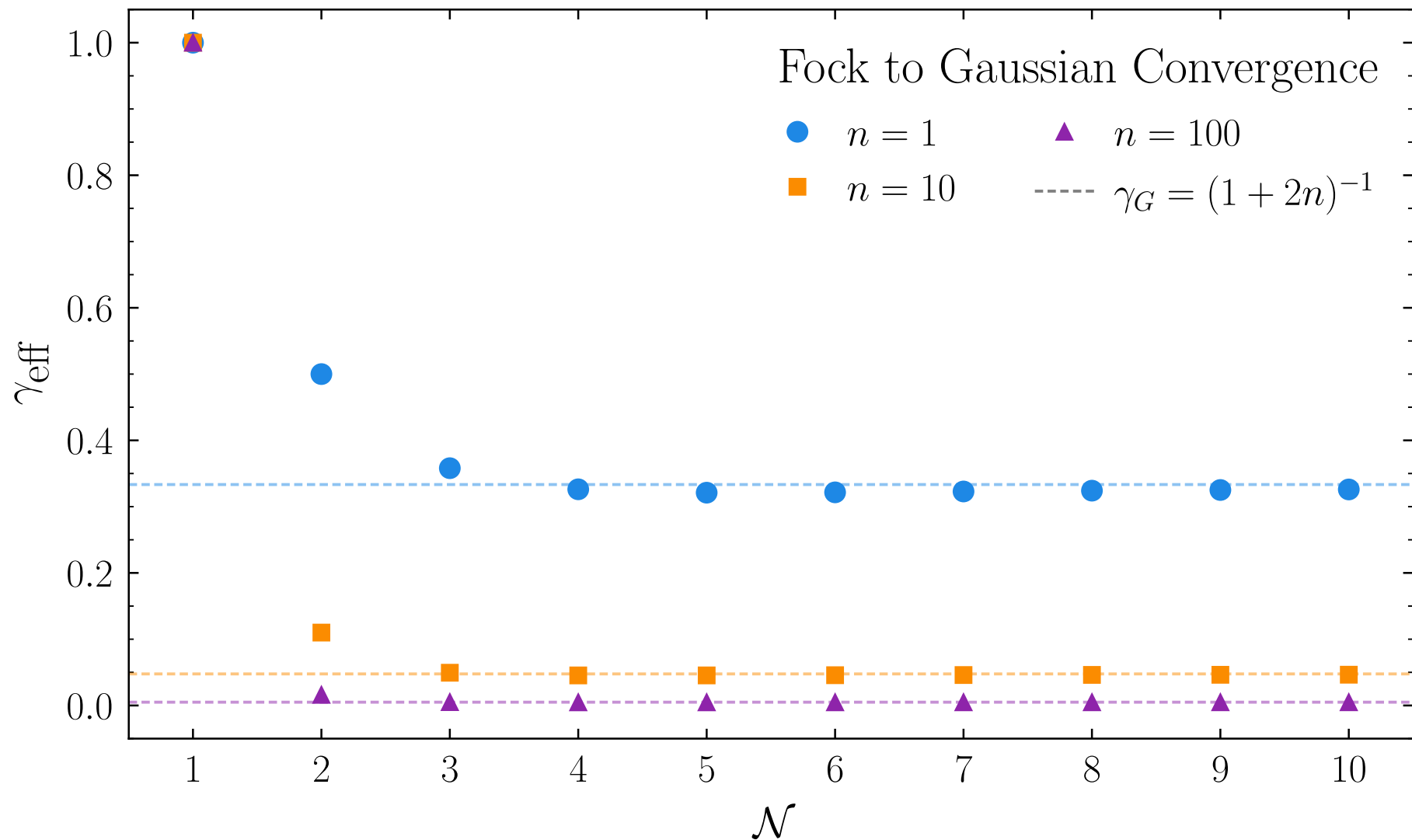
$$P_{\text{eff}}(\alpha) = \int \left(\prod_i d^2\alpha_i \right) P(\alpha_1, \alpha_2, \dots) \delta\left(\alpha - \sum_i C_{\mathbf{p}} e^{-iK_{\mathbf{p}}t} \alpha_i\right)$$

If the plane wave modes are independent, $P(\alpha_1, \alpha_2, \dots) = \prod_i P_i(\alpha_i)$

By central limit theorem, this convolution yields Gaussian state $P_{\text{eff}}(\alpha) \propto e^{-|\alpha|^2/\langle n \rangle}$

Makes axion behave like a classical Gaussian random field,
regardless of state of the fundamental plane wave modes

Quantum Central Limit Theorem



Convergence of central limit theorem rapid, even for high mode occupancy

However, can evade theorem with mode correlations, nonstationary axions, Bose-Einstein condensation, or laboratory-sourced axions

For now, will let the effective mode's quantum state be arbitrary

Evolution of the Cavity State

For simplicity, projectively measure cavity after short time t_m , where $K_p t_m \lesssim 1$

Straightforward to compute the time evolution in general

$$P_{\text{cav}}^f(\alpha) \simeq \int d^2\beta \frac{P_{\text{eff}}(\beta/\sqrt{\eta})}{\eta} P_{\text{cav}}^i(\alpha - \beta) = \frac{P_{\text{eff}}(\alpha/\sqrt{\eta})}{\eta}$$

scaled convolution of axion
state and initial cavity state

take initial cavity vacuum state
for now, just for simplicity

Conversion efficiency at maximum $t_m \sim Q_c/m_a \sim 10^{-4}$ s

$$\eta \simeq g^2 t_m^2 \sim 10^{-21} \left(\frac{g_{a\gamma\gamma}}{10^{-15} \text{ GeV}^{-1}} \frac{B_0}{10 \text{ T}} \frac{Q_c}{10^5} \frac{10^{-5} \text{ eV}}{m_a} \right)^2$$

Any negativity in P_{eff} directly imprinted on cavity state, but scaled by small η

Example: Nonclassical Number Statistics

Cavity number distribution $p_n = \int d\alpha P_{\text{eff}}(\alpha) |\langle n | \sqrt{\eta} \alpha \rangle|^2 = \sum_{k=n}^{\infty} \binom{k}{n} p_k^{\text{DM}} \eta^n (1 - \eta)^{k-n}$

Note that $\text{var}(n) - \langle n \rangle = \int d\alpha P(\alpha) (|\alpha|^2 - \langle n \rangle)^2 \geq 0$ if $P(\alpha) \geq 0$

Thus, only nonclassical states have Mandel $Q = (\text{var}(n)/\langle n \rangle) - 1 < 0$

DM coherent state: $Q_{\text{DM}} = 0$, Poisson number distribution

DM Gaussian state: $Q_{\text{DM}} = N_{\text{eff}}$, geometric number distribution

DM Fock state: $Q_{\text{DM}} = -1$, definite number

But the value of Q imprinted in the cavity state is suppressed, $Q_{\text{cav}} = \eta Q_{\text{DM}} \geq -\eta$

Intuitively, because converting axions to photons itself
imprints approximately Poisson fluctuations

Distinguishing Number Statistics

Consider edge of sensitivity, $p_2 \ll p_1 \ll p_0 \approx 1$, mean signal photons $\eta N_{\text{eff}} \simeq p_1 \sim 10^{-2}$

Time to discover axion at this coupling: t_m/p_1 (reasonable time by definition)

$$\text{Infer Mandel } Q \text{ through its effect on } p_2 \simeq \begin{cases} p_1^2/2 & \text{DM coherent state} \\ p_1^2 & \text{DM Gaussian state} \\ p_1(p_1 - \eta)/2 & \text{DM Fock state} \end{cases}$$

Time to distinguish coherent and Gaussian: t_m/p_1^2 (reasonable for post-discovery)

$$\text{Time to see negative } Q: t_m/\eta^2 \sim 10^{30} \text{ years} \left(\frac{10^{-15} \text{ GeV}^{-1}}{g_{a\gamma\gamma}} \frac{10 \text{ T}}{B_0} \right)^4 \left(\frac{10^5}{Q_c} \frac{m_a}{10^{-5} \text{ eV}} \right)^3$$

Quadrature Statistics

Simplest nonclassicality measure for quadrature measurements is squeezing

$$X = (a + a^\dagger)/\sqrt{2} \quad S = \text{var}(X) - 1/2$$

DM coherent state: $S_{\text{DM}} = 0$

DM Gaussian state: $S_{\text{DM}} = N_{\text{eff}}$ (all have Gaussian quadrature distribution)

DM squeezed state: $S_{\text{DM}} < 0$

Starting from an initial vacuum state, DM can induce squeezing $S_{\text{cav}} = \eta S_{\text{DM}}$

Time to distinguish coherent and Gaussian: $t_m/(\eta N_{\text{eff}})$ (reasonable)

Time to detect negative S_{cav} : t_m/η^2 (impractically long)

Similar story for higher-order generalizations of Q and S

Entanglement and Decoherence

Entangled effective modes can transmit entanglement to cavity modes

Classical correlations can be large,
but entanglement measures are not

$$E = \frac{\text{var}(X_1 - X_2) + \text{var}(Y_1 + Y_2)}{2} - 1 \geq -1$$
$$E_{\text{cav}} = \eta E_{\text{DM}} \geq -\eta$$

Axion-cavity interaction can decohere a prepared initial cavity state

Quantify by the purity loss $\delta = 1 - \text{tr}(\rho_c^2)$

$$\frac{\delta}{2\eta} = \left(\langle a^\dagger a \rangle - |\langle a \rangle|^2 + \frac{1}{2} \right) \left(\langle c^\dagger c \rangle - |\langle c \rangle|^2 + \frac{1}{2} \right) - \frac{1}{4} - (\langle (a^\dagger)^2 \rangle - \langle a^\dagger \rangle^2) (\langle c^2 \rangle - \langle c \rangle^2) + \text{c.c.}$$

Lowest for coherent DM, $\delta \sim \eta$, highest for Gaussian DM, $\delta \sim \eta N_{\text{eff}}$

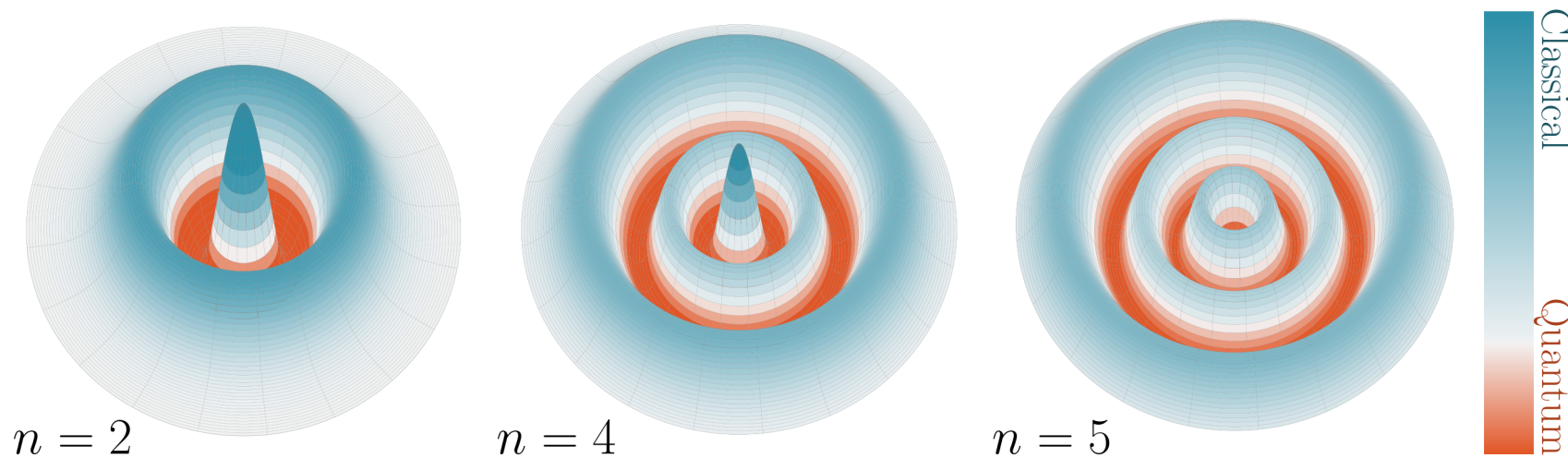
Nonclassical axion states extend the range up and down, by amount suppressed by η

A Heuristic Geometric Argument

The probability $Q(\alpha)$ to be in a coherent state $|\alpha\rangle$ can't be negative:

$$Q(\alpha) = \langle \alpha | \rho | \alpha \rangle = \int d\beta P(\beta) |\langle \beta | \alpha \rangle|^2 = \int d\beta P(\beta) e^{-|\alpha-\beta|^2} \geq 0$$

Thus, **any** negative P -function must effectively oscillate on $\mathcal{O}(1)$ scales



For initial vacuum state,
 $P_{\text{cav}}^f(\alpha) = P_{\text{eff}}(\alpha/\sqrt{\eta})/\eta$ oscillates on
 $\sqrt{\eta}$ scales, negativity hard to observe

Thermal fluctuations in the initial state also convolve final state with a Gaussian!

This will already remove negativity unless $T \lesssim \frac{m_a}{\log(1/\eta)} \simeq 2 \text{ mK} \left(\frac{m_a}{10^{-5} \text{ eV}} \right)$

A Precise General Argument

For any nonclassical DM state, construct the classical state $P'_{\text{DM}}(\alpha) = Q_{\text{DM}}(\alpha) \geq 0$

Probabilities of a generic measurement outcome in both cases:

$$p = \int d\alpha P_{\text{cav}}^f(\alpha) f(\alpha) \quad p' = \int d\alpha P_{\text{cav}}^f(\alpha) \int d\beta f(\alpha - \beta) \frac{e^{-|\beta|^2/\eta}}{\pi\eta}$$

$$p - p' \lesssim \text{tr} |\rho^f - \rho_{\text{cl}}^f| \lesssim \eta \text{tr} |\mathcal{L}[\rho^f]| \quad \begin{array}{l} \text{difference in the probability always} \\ \text{suppressed by } \eta \end{array}$$

To cancel the suppression, the initial detector state must be nonclassical, **and** properties of the initial state must scale as $1/\eta$

$$\text{var}(X_c) = (1 - \eta) \text{var}(X_{c,0}) + \eta \text{var}(X_a)$$

Such as squeezed state with occupancy $\sim 1/\eta$, far beyond current technology

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Detectors only see coarse-grained modes, not fundamental modes

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- Addressing the literature

The Expected Axion State

The axion appears classical because it has high occupancy

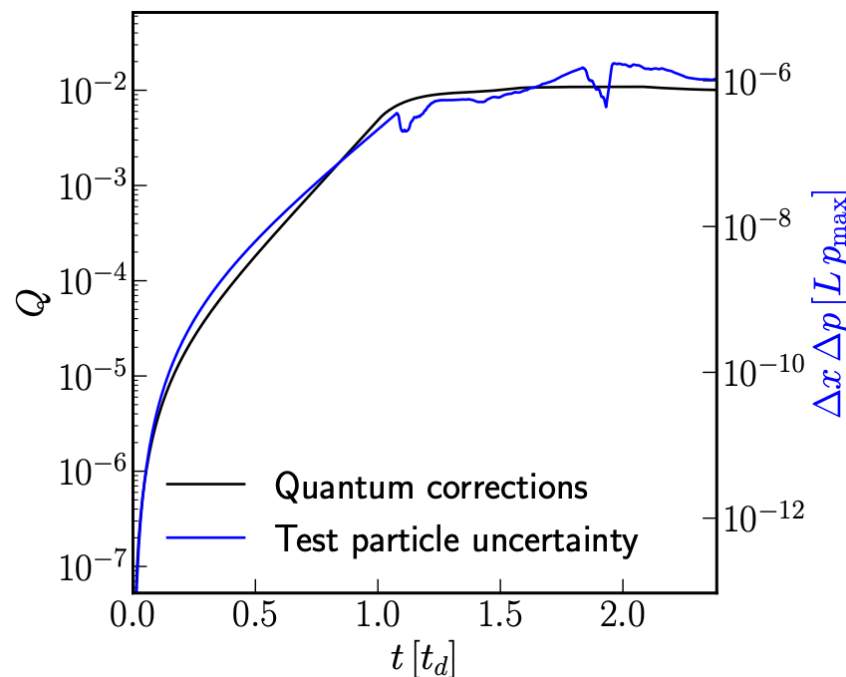
Misalignment produces the axion in a coherent state

Usual statement in papers, but not real reason

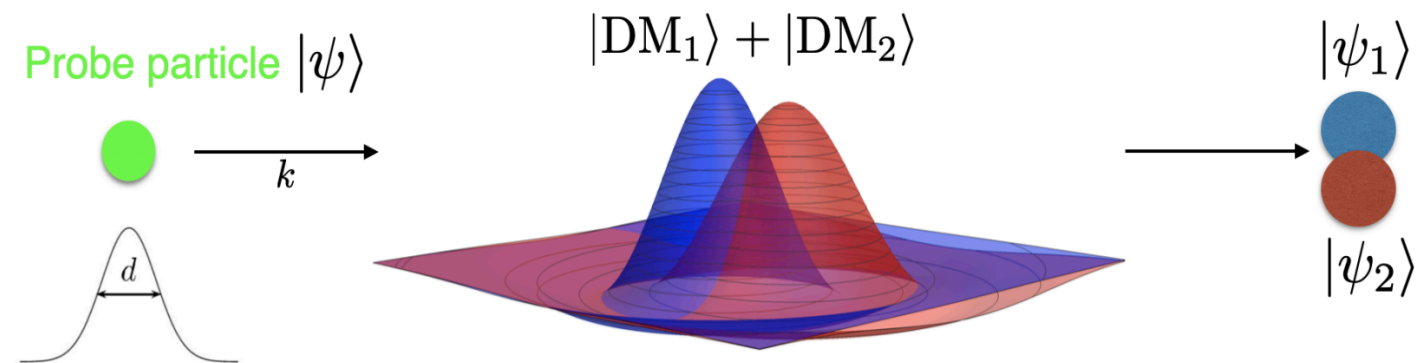
Only produces “coherently oscillating” field
 $(\Delta\omega_a \ll m_a)$

Inflation produces two-mode squeezed states

Virialization likely drives to a Gaussian state



Eberhardt et al., PRD (2024)

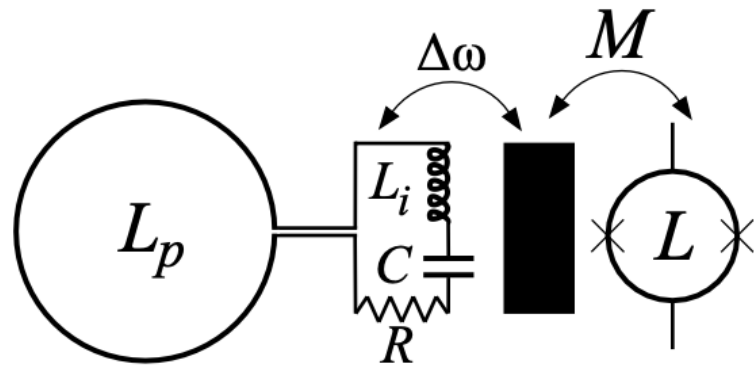


Allali and Hertzberg, PRL (2021)

True state uncertain: quantum spread grows rapidly during halo collapse, and superpositions like $|\alpha\rangle + |-\alpha\rangle$ have negligible gravitational decoherence

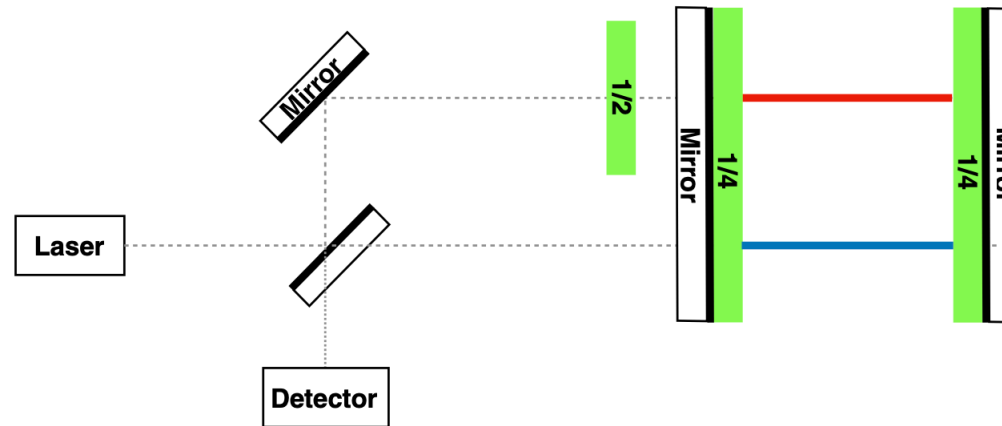
Extension to Other Detectors

Our results apply to almost all axion DM detectors, as their interactions can be written in a linear mixing form



lumped element circuits

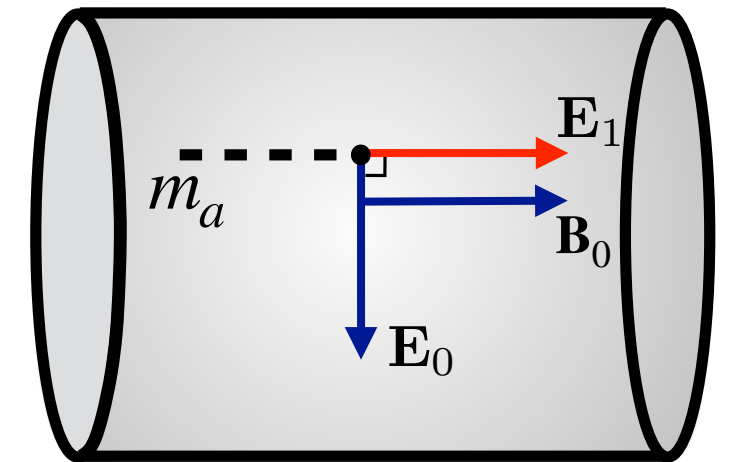
Kahn, Safdi, Thaler, PRL (2016)



axion interferometry

DeRocco and Hook, PRD (2018)

$$H_{\text{int}} = ig (c^\dagger a - c a^\dagger)$$



superconducting heterodyne

Berlin, D'Agnolo, Ellis, Nantista, Neilson, Schuster, Tantawi, Toro, KZ, JHEP (2020)

Also holds for many GW detectors, such as laser interferometers and Weber bars

$$\eta \sim \frac{B_0^2 L^2}{M_{\text{pl}}^2} \sim 10^{-33} \left(\frac{B_0}{10 \text{ T}} \right)^2 \left(\frac{L}{10 \text{ m}} \right)^2$$

graviton-photon conversion

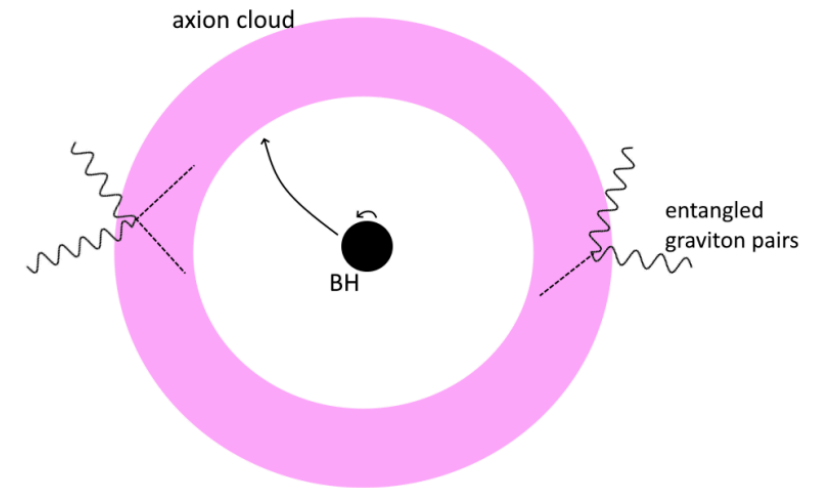
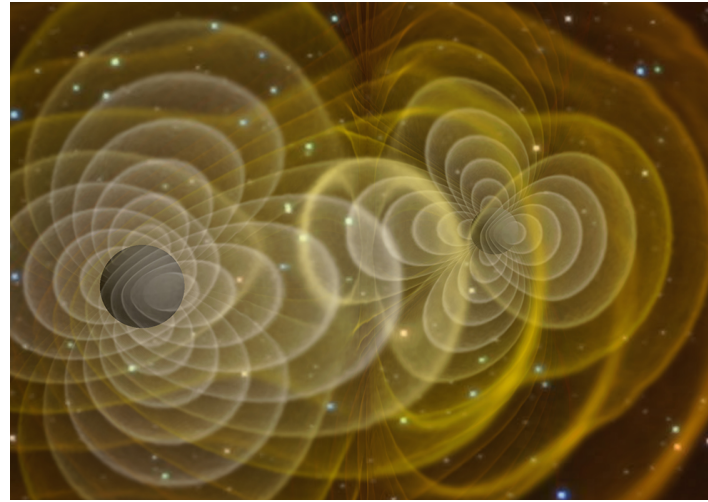
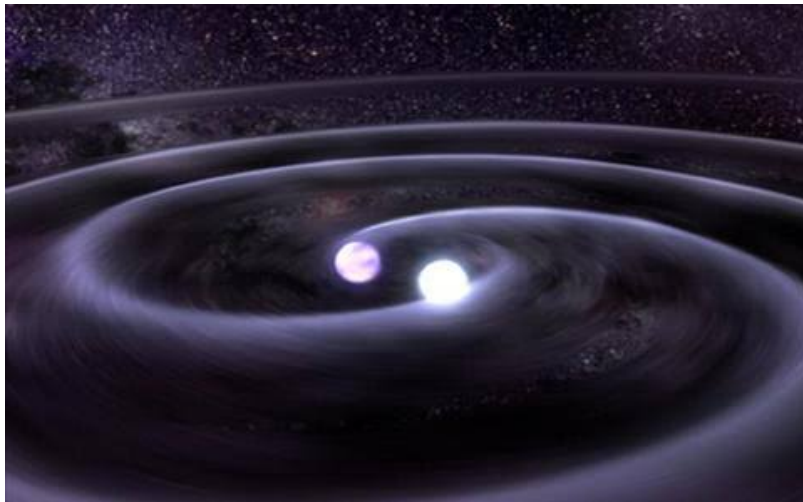
$$\eta \sim \frac{\rho Q_m v_s^5}{\omega_0^2 M_{\text{pl}}^2} \sim 10^{-32}$$

Weber bar

The Expected Graviton State

Gravitational waves are produced in a coherent state

Textbook statement; only holds for linearized gravity and a perfectly known source



Nonlinear interactions in a binary inspiral can make a (weakly) squeezed state

Kanno, Soda, Taniguchi, PRL (2026)

(also realized indirectly in amplitudes literature)

Britto, Gonzo, Jehu, JHEP (2022)

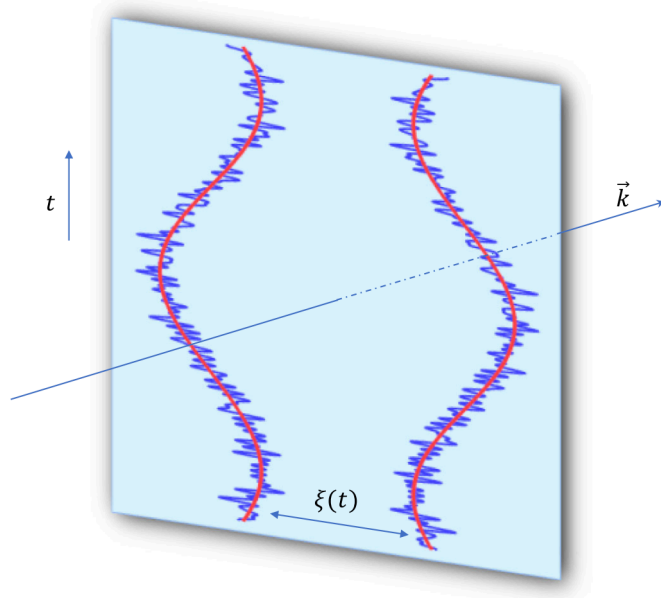
For stochastic gravitational wave background, expect Gaussian state

Superradiant axion clouds could source strongly squeezed GWs

Dorlis et al., PRL (2026)

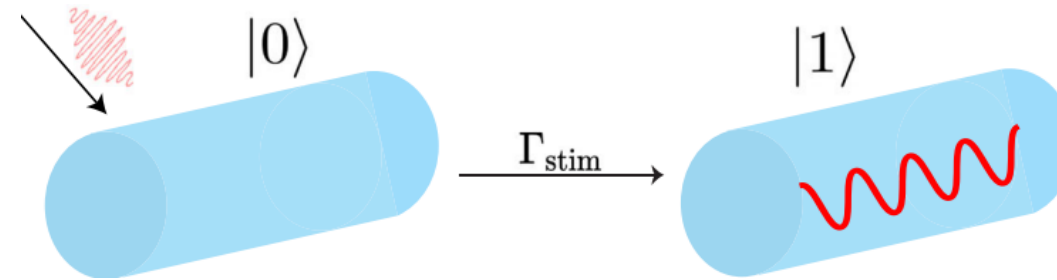
(also squeezed gravitons from inflation)

What About Quantum Gravity Claims?



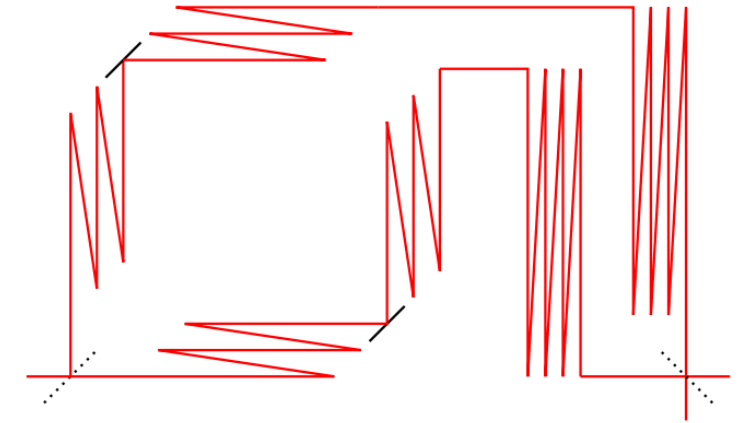
“The noise of gravitons”

Parikh, Wilczek, Zahariade, PRL (2021)



Absorption of gravitons

Tobar et al., Nature Commun. (2024)



State-dependent decoherence

Schutzhold, PRL (2025)

Classical GWs can yield
 $\text{var}(X) \gg 1/2$

Classical GWs can excite
discrete energy levels

Classical GWs can yield
low/high decoherence

$\text{var}(X) < 1/2$ would be
quantum, can't see

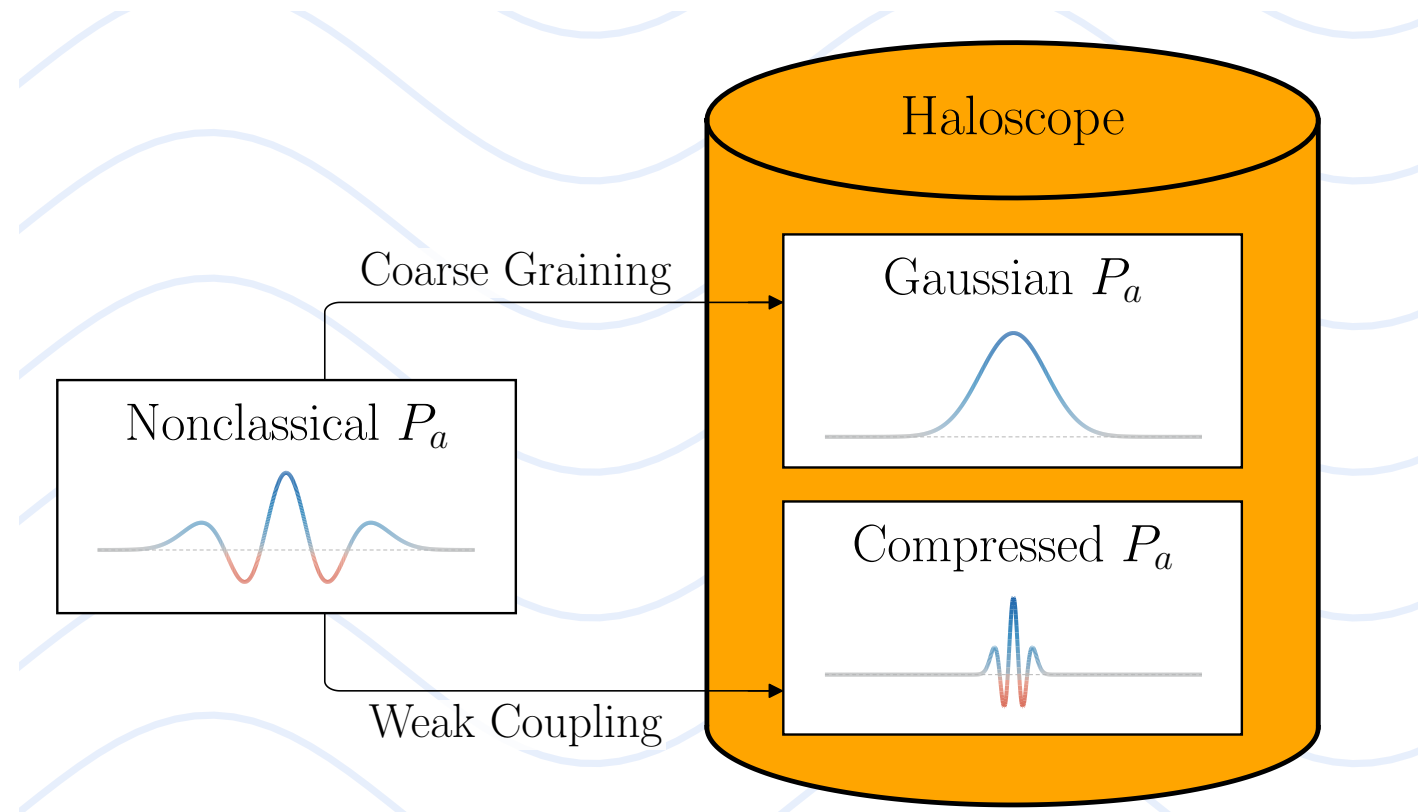
$Q < 0$ would be
quantum, can't see

Extended range would
be quantum, can't see

Other works drop the η suppression, or consider quantity that isn't intrinsically quantum, or compare only to a coherent state rather than a general classical ensemble

Conclusion

Weakly coupled waves, such as axion DM and GWs, act approximately classically



Not because of high occupancy, but rather by coarse-graining and weak coupling

The classical field approximation works, and realistic DM and GW detectors cannot show the axion/gravitational field is quantized